

Canton Cortex Explorer: Operational Intelligence and Neural Navigation Guide

Canton Cortex Explorer is a hybrid analysis terminal that transforms static blockchain ledgers into a living, breathing, and real time responsive **3D Neural Topology** simulation. This document explains all layers of the system, its analysis algorithms, and operational capacities in full detail.

1. Introduction: "Seeing the Mind of the Blockchain"

While traditional explorers answer the question "what happened?", Canton Cortex Explorer focuses on "where and why did it happen?". Thanks to the central data engine on *useCityStore.ts* and deterministic algorithms within *cityStore.helpers.ts*, the system transforms raw hash strings into spatial entities.

2. Operational Launch: Neural Uplink Sequence

The process of accessing the system initiates a military grade data decryption sequence known as the **Neural Uplink**.

A. Authentication (OIDC Handshake)

Managed through *page.tsx* and *useLedger.ts*, the following steps are triggered when a **Target Party ID** is entered:

1. **Auth Request:** A Bearer Token is requested via the OIDC protocol through the MODO API.
2. **Identity Verification:** The actAs permissions within the token are decrypted to stabilize the neural connection.
3. **Neural Log Terminal:** When the *DECRYPTING_NEURAL_DATA...* message appears in the left terminal, "Artifacts" from the ledger begin to be captured.

B. Topological Reconstruction

Once data is collected, the finishScan function takes over. The city is not built randomly; it is reconstructed in **Batches** (groups of 40) based on interaction strength. This ensures a smooth launch by balancing CPU/GPU loads.

3. Neural City Architecture (Reading the Topology)

Every architectural element in the city is a physical projection of a data set.

A. District Architecture

Buildings are separated into neighborhoods based on ClusterHint types and EntityClass categorizations:

- **APP_DISTRICT (Green):** Smart contracts and DApp interactions.
- **GOV_DISTRICT (Purple):** Governance, voting, and protocol decisions.
- **VAL_DISTRICT (Yellow):** Validator nodes providing network infrastructure.
- **OPS_DISTRICT (Blue):** Operational and high volume administrative flows.
- **GENERAL:** Independent participants that cannot be included in a specific cluster.

B. Building (Node) Anatomy

Physical properties of buildings are determined by the `getRelationWeight` function in `cityStore.helpers.ts`:

- **Height:** A weighted average of transaction count (log10), asset holdings, and recent activity score.
- **Pulse:** Buildings that have transacted within the last 5 minutes increase their `emissiveIntensity`, causing the building to "breathe."
- **Types:**
 - **CENTER (Blue Dome):** The main monitored wallet. Protected by the `PrivacyDome`.
 - **SUPER_VALIDATOR:** High security nodes surrounded by golden rings.
 - **PARTICIPANT:** Standard data points.

C. Privacy Dome

The blue dome at the center of the city resonates based on the wallet's **Net Flow** status:

- **Green Glow:** Inbound flow (Money entering).
- **Pink Glow:** Outbound flow (Money exiting).
- **Red Jitter:** Critical Error or Risk Signal.

4. Data Flow Dynamics: Neon Paths

Lasers flying between buildings represent value transfers on the blockchain.

- **Plasma Balls:** The size of the light balls within the laser grows logarithmically based on the transfer **Amount**.
- **Flow Speed:** Plasma balls accelerate based on the recency of the transaction.
- **Whale Alert:** For transactions over 1000 CC, a holographic Database icon appears on the path.
- **Red Lines:** Flows marked as risky by the system (Risk Detected) are shown in red neon.

5. Intelligence Overlay

The Explorer is more than a simple viewer; it is an active intelligence tool.

A. Deterministic Risk Analysis (Risk Flags)

The system automatically detects the following conditions using the rule set in `cityStore.helpers.ts`:

- **SELF_TRANSFER:** Circular transactions sent by the wallet to itself.
- **HIGH_VALUE_TRANSFER:** Unusually high volume flows.
- **ONE_WAY_FLOW:** One way flows that raise suspicion of liquidity draining.
- **LOW_CONFIDENCE:** Suspicious links with low decryption quality.

B. Relationship Scoring

Every connection receives a score between 0-100:

- **Volume + Frequency:** Transaction frequency and magnitude.
- **Recency:** Interaction within the last 24 hours grants the highest points.
- **Confidence:** The technical verification level of the connection.

6. Operational Tools and Dashboard

A. Global Stats (Top Bar)

Presents the network's biometric data:

- **TPS Density:** Real time transaction density per second.
- **Rel_Strength:** The ratio of High Confidence (HC) and validator interactions.
- **Amulet Value:** Live value of the network's native asset.

B. Tactical HUD (Left Panel)

Deciphers micro data about the focused target. Can be collapsed using the `isLeftPanelCollapsed` feature.

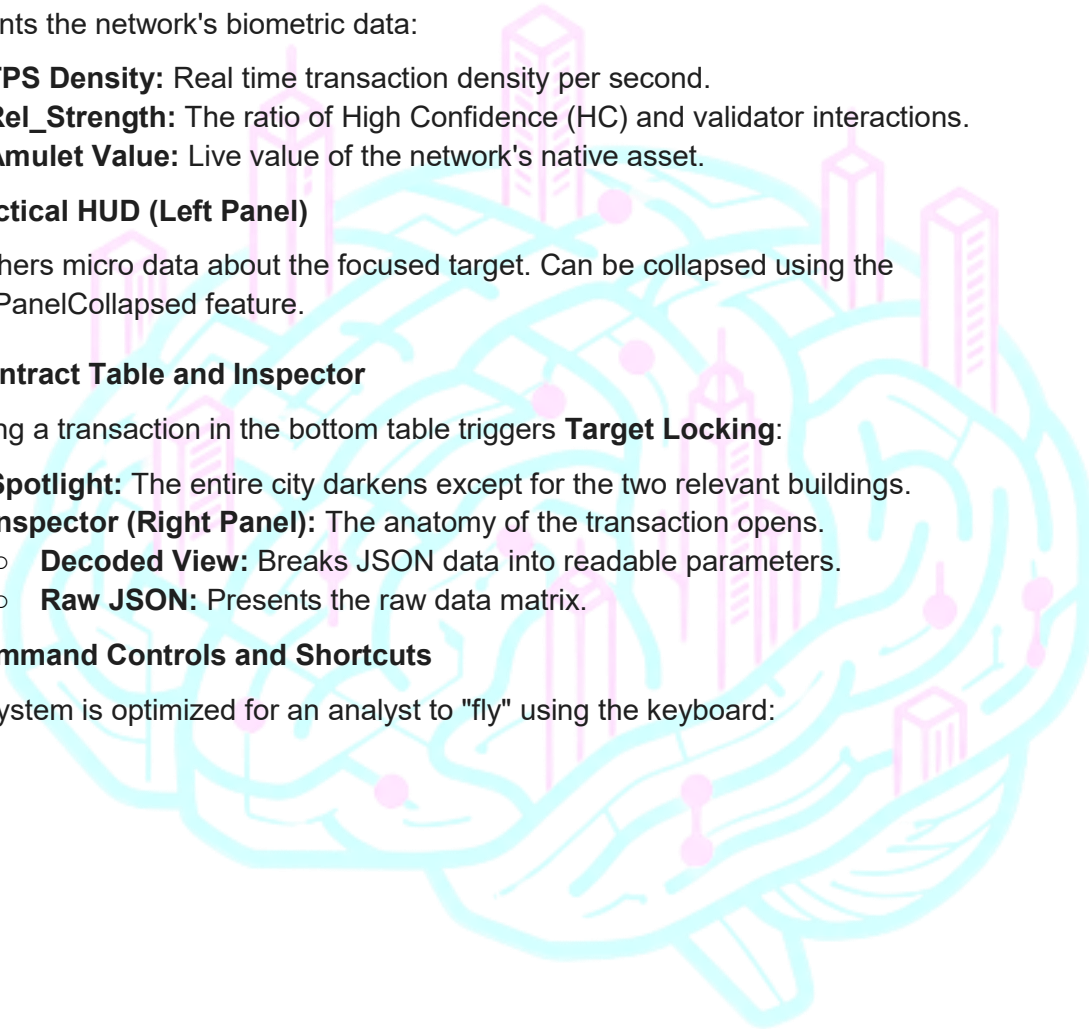
C. Contract Table and Inspector

Clicking a transaction in the bottom table triggers **Target Locking**:

1. **Spotlight:** The entire city darkens except for the two relevant buildings.
2. **Inspector (Right Panel):** The anatomy of the transaction opens.
 - **Decoded View:** Breaks JSON data into readable parameters.
 - **Raw JSON:** Presents the raw data matrix.

7. Command Controls and Shortcuts

The system is optimized for an analyst to "fly" using the keyboard:



Key	Function	Technical Description
[H]	HUD Toggle	Hides all 2D layers, switching to pure 3D vision mode.
[I]	Intel Summary	Opens the holistic X ray of the relational network (Cluster Districts).
[ESC]	Abort/Close	Instantly closes Inspector or Intel panels, clears selection.
[CTRL + L]	Clear Logs	Clears terminal logs on the login screen.
[Left Click]	Interact	Used to select buildings or table rows.
[Right Click]	Pan	Performs panning in the 3D scene.

8. Security and Warning Protocols

- **ISOLATED UPLINK:** Indicates that data access is sealed per Canton Privacy Protocols.
- **VERIFIED:** A blue shield icon confirms the node is a system verified institution or validator.
- **GLITCH CORE:** Visual jitter effects occur on buildings during momentary synchronization loss or high Jitter in the network.

Canton Cortex Explorer: Designed to manage the complex mind of the blockchain and transform data into intelligence.